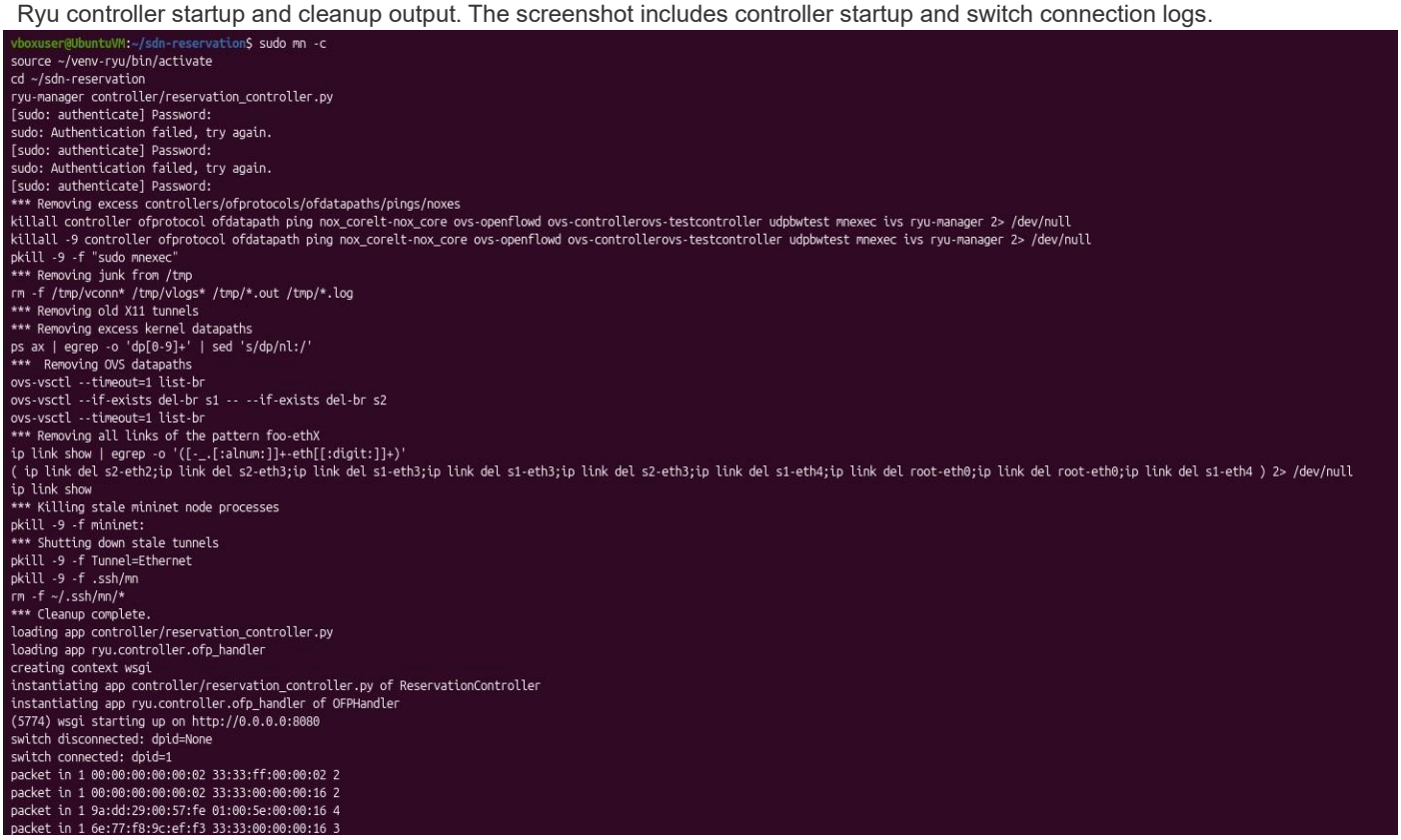
Network Resource Reservation System

Review 2 — Implementation and Testing Evidence.

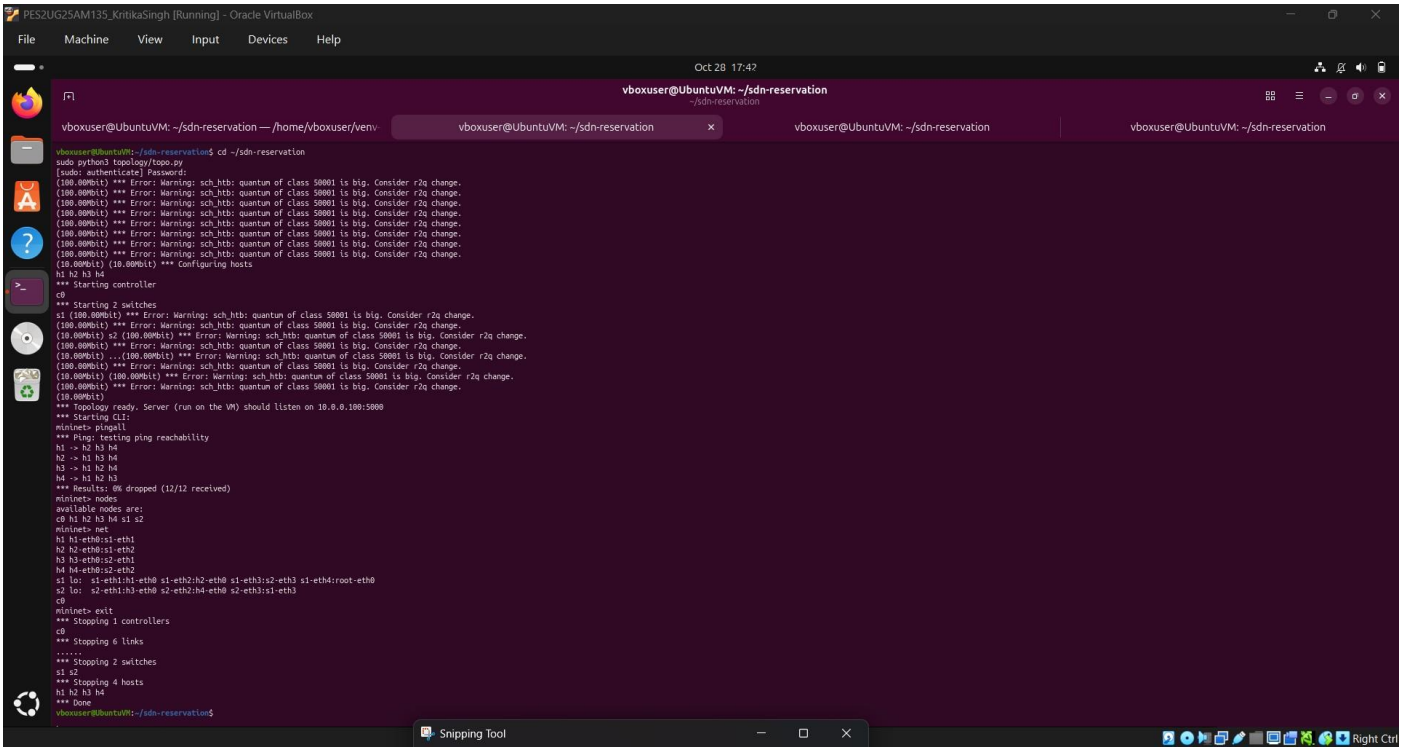
1. SDN Controller Startup

Ryu controller startup and cleanup output. The screenshot includes controller startup and switch connection logs.



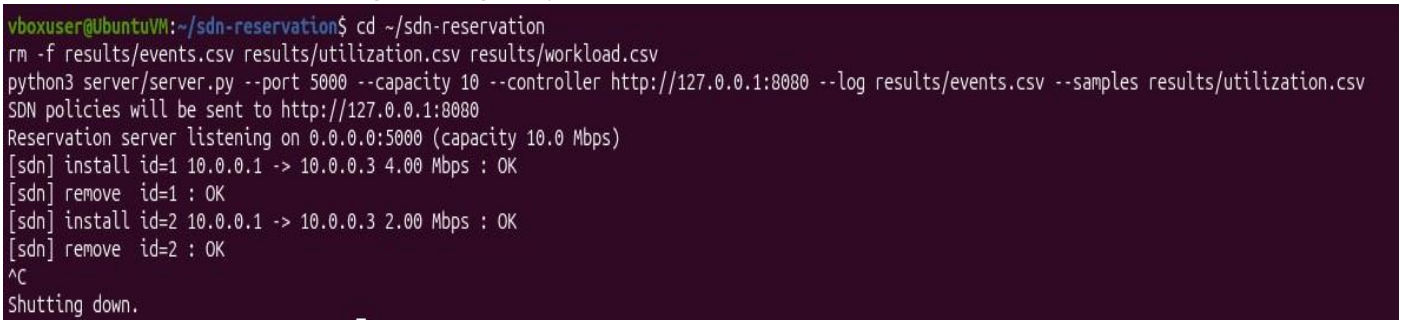
2. Mininet Topology and Connectivity

Custom Mininet topology with four hosts and two switches; pingall reports 0% dropped (12/12 received). The nodes and network links are also shown.



3. TCP Reservation Server and SDN Policy Logs

Reservation server startup and logs showing policy installation and removal events.



4. Reservation Acceptance, Rejection, Status, and List

Client outputs show an accepted 4 Mbps reservation, a rejected 7 Mbps request due to insufficient bandwidth, status for reservation ID 1, and the reservation list.

```
vboxuser@UbuntuVM:~/sdn-reservation$ python3 client/client.py reserve 10.0.0.1 10.0.0.3 4 120
{
  "status": "ACCEPTED",
  "id": 1,
  "src": "10.0.0.1",
  "dst": "10.0.0.3",
  "bandwidth_mbps": 4.0,
  "duration_s": 120.0
}
vboxuser@UbuntuVM:~/sdn-reservation$ python3 client/client.py reserve 10.0.0.2 10.0.0.4 7 120
{
  "status": "REJECTED",
  "reason": "INSUFFICIENT_BANDWIDTH",
  "message": "not enough free bandwidth on the link",
  "available_mbps": 6.0
}
vboxuser@UbuntuVM:~/sdn-reservation$ python3 client/client.py status 1
{
  "status": "OK",
  "id": 1,
  "src": "10.0.0.1",
  "dst": "10.0.0.3",
  "bandwidth_mbps": 4.0,
  "remaining_s": 83.2
}
vboxuser@UbuntuVM:~/sdn-reservation$ python3 client/client.py list
{
  "status": "OK",
  "reservations": [
    {
      "id": 1,
      "src": "10.0.0.1",
      "dst": "10.0.0.3",
      "bandwidth_mbps": 4.0,
      "remaining_s": 65.1
    }
  ]
}
```

5. Reservation Release and Policy Removal

The client releases reservation ID 1, and the controller policy endpoint returns an empty policy list with the best-effort limiter disabled.



6. Resource Utilization, Events, and Expiry

CSV excerpts show utilization and reservation events. A 2 Mbps, 15-second reservation is accepted, and a subsequent list output is empty after expiry.

```
vboxuser@UbuntuVM:~/sdn-reservation$ head -n 15 results/utilization.csv
time,elapsed_s,used_mbps,utilization_pct,active_reservations
1793208235.375,1.0,0.00,0.0,0
1793208236.376,2.0,0.00,0.0,0
1793208237.377,3.0,0.00,0.0,0
1793208238.378,4.0,0.00,0.0,0
1793208239.380,5.0,0.00,0.0,0
1793208240.381,6.0,0.00,0.0,0
1793208241.381,7.0,0.00,0.0,0
1793208242.383,8.0,0.00,0.0,0
1793208243.384,9.0,0.00,0.0,0
1793208244.385,10.0,0.00,0.0,0
1793208245.386,11.0,0.00,0.0,0
1793208246.389,12.0,0.00,0.0,0
1793208247.389,13.0,0.00,0.0,0
1793208248.391,14.0,0.00,0.0,0
vboxuser@UbuntuVM:~/sdn-reservation$ cat results/events.csv
time,event,id,src,dst,bandwidth_mbps,used_mbps,utilization_pct,reason
1793208542.132,ACCEPTED,1,10.0.0.1,10.0.0.3,4.0,4.00,40.0,
1793208554.047,REJECTED,,10.0.0.2,10.0.0.4,7.0,4.00,40.0,INSUFFICIENT_BANDWIDTH
1793208625.419,RELEASED,1,10.0.0.1,10.0.0.3,4.0,0.00,0.0,
vboxuser@UbuntuVM:~/sdn-reservation$ python3 client/client.py reserve 10.0.0.1 10.0.0.3 2 15
{
  "status": "ACCEPTED",
  "id": 2,
  "src": "10.0.0.1",
  "dst": "10.0.0.3",
  "bandwidth_mbps": 2.0,
  "duration_s": 15.0
}
vboxuser@UbuntuVM:~/sdn-reservation$ python3 client/client.py list
{
  "status": "OK",
  "reservations": []
}
```